DOCUMENT 2

Product Requirements Document (PRD)

2.1 Product summary

StayZen is a web platform with three portals. **Guests** search hotels by district or GPS, see them ranked by a Best Match Score, compare options and send a pre-booking request (no online payment) that returns a Booking ID. With that Booking ID they open a **live crowd heat map** of the hotel's facilities, see wait-time ranges, and after check-in take a free virtual queue token. **Hotel managers** register their hotel, accept bookings and type facility head counts. The **Stays Admin** verifies hotels before they can be booked. This answers MRD needs N1–N5 (section 2.6).

2.2 Features and MoSCoW priority

ID	FEATURE (CAPABILITY)	DESCRIPTION	PRIORITY	MRD NEED
F1	**Hotel discovery with Best Match Score**	Search by district or GPS, budget, guests, dates and wanted facilities. Each hotel gets a score out of 100: budget fit 40 + rating 25 + distance 20 + facilities 15. Results in pages of 10 with Previous/Next.	MUST	N1
F2	**Pre-booking with Booking ID**	Guest sends a request for a verified hotel; system calculates total (nights × price × rooms) and returns a unique ID (SZ-1234). Hotel accepts or rejects. No payment.	MUST	N2
F3	**Live facility crowd heat map**	Staff type head counts; guest enters their Booking ID to see a floor-plan heat map (Low <50%, Medium 50–79%, High ≥80%), a crowded alert, a wait-time range, and a "may be outdated" warning after 15 minutes.	MUST	N3
F4	Hotel comparison	Compare up to 3 hotels side by side; best one highlighted; ties broken by distance, price and facilities.	SHOULD	N1
F5	Hotel details page	Rooms and prices, facility details (open hours, trainer, equipment, menu), parking (watchman, car and bike slots), Wi-Fi, AC rooms, food within 1 km, contact, route.	SHOULD	N1, N4
F6	Virtual queue token and activity suggestions	After check-in, a free token (Q-017) for a crowded facility with queue position; staff "Call next"; suggestions of quiet facilities that fit the wait.	SHOULD	N3
F7	Hotel registration and admin verification	Manager signs up with their hotel (pending); admin verifies, rejects, suspends; only verified hotels accept pre-bookings; every edit logged for review; outdated crowd data flagged.	SHOULD	N4

F8	StayZen assistant (English and Tamil, text and voice)	Chat about crowds, directions, tokens, prices and the website; answers hotel facts only from StayZen data; says "needs confirmation" when data is missing.	COULD	N5
F9	Safety and emergency help	Emergency numbers (112, 100, 108, 101) and one-tap navigation to the nearest police station, hospital and pharmacy.	COULD	N5
F10	Finish trip and feedback	Guest marks the trip complete and rates 8 areas (cleanliness, crowd accuracy, queue experience…); admin sees averages.	COULD	N4
F11	Online payment	Pay for rooms inside StayZen.	WON'T	–
F12	Camera / sensor / Wi-Fi based automatic counting	Automatic people counting from hardware.	WON'T	–

Must-Haves are capped at three (F1–F3). Everything else is built only after the Must-Haves pass their acceptance criteria.

2.3 Success metrics (measurable in the demo)

ID	METRIC	TARGET	HOW WE MEASURE IT IN THE DEMO
M1	Time for a guest to go from login to a Booking ID	Under 2 minutes, 6 clicks or fewer	Stopwatch and click count during dry run (QA)
M2	Time for staff to update one facility head count	Under 10 seconds	Stopwatch on the Facility status screen
M3	Delay before a staff update appears on the guest heat map	60 seconds or less	Update count, watch guest screen (auto-refresh 60 s or Refresh)
M4	Crowded facilities (≥80%) that show an alert and a wait-time range	100%	Test cases TC-05 to TC-07 in the QA sheet

2.4 Primary user flows (Must-Have features)

F1 + F2 · Find and pre-book a hotel (Priya)

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Sign in (Guest) → enter district, budget, guests, dates, wanted facilities → Find hotels
→ list ranked by Best Match Score (Previous / Next pages) → click a hotel → hotel page
→ Pre-book → confirm total → Booking ID SZ-xxxx shown (status: Pending)
→ Hotel Management: Pre-bookings → Accept → guest sees "Confirmed"
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F3 · Check the crowd before going (Priya and Ravi)

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Hotel Management: Facility status → type head count → Save (last-updated time stored)
Guest: Crowd map → enter Booking ID → floor-plan heat map + "Crowd right now" bars
→ tap Gym (High 85%) → "This facility is crowded" + wait 15–20 min
→ (after check-in) Get free virtual token → Q-017, position 3
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2.5 Out of scope (explicit)

- Online payment, refunds and cancellation charges.
- Cameras, sensors, Wi-Fi router counting or any hardware.

- Trained machine-learning crowd prediction (we show current counts and rule-based ranges only).
- Turn-by-turn indoor navigation (we show a simple floor plan, route line and text directions).
- "Continue with Google" sign-in (needs Google OAuth setup; email sign-in is used).
- Real prices and facility details for OpenStreetMap hotels until their owners register (sample values are used in the prototype).

2.6 MRD needs referenced

ID	NEED (FROM MRD)	PERSONA
N1	Choose the hotel that truly fits my budget, distance and facilities, quickly	Priya, Arun
N2	Hold a room with confidence without paying an unknown hotel online	Priya, Arun
N3	Know which facility is crowded and how long I would wait, before walking there	Priya, Ravi
N4	Trust that the hotel and its information are real and up to date	Priya, Stays Admin
N5	Get help in my own language, including in an emergency	Arun, Priya